RATIONAL TAILS PERSIST UNDER FIELD EXTENSION AS DISJOINT UNIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the intersection compatibility X_K cap C_i = x_i is the base change of the defining property X cap C = {x} from Example example-rational-tail; the decomposition of C_K uses the standard base change of projective space along k' tensor_k K -> product of the K'_i; dependencies carried as navigation stubs (Example example-rational-tail, Proposition proposition-projective-line).

Editorial note. The label lemma-rational-tail-field-extension names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Defining data of a rational tail. Same unit, Example example-rational-tail. The proof uses the point x in C with residue field k' = kappa(x) supplied by the definition of a rational tail, and the defining property X cap C = {x}.
- Projective line identification. Same unit, Proposition proposition-projective-line. The proof uses C cong P^1_{k'}, given by Proposition proposition-projective-line.

Source reference. Original reference 0E3H: sources/stacks-curves/pinned-original-source.tex, source lines 5933–5933. Prerequisite pointer invoked at proof line(s) 6014; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0C6U: sources/stacks-curves/pinned-original-source.tex, source lines 2138–2138. Prerequisite pointer invoked at proof line(s) 6022; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be a field. Let X be a proper scheme over k having dimension 1 and $H^0(X, \mathcal{O}_X) = k$. Assume the singularities of X are at-worst-nodal. Let $C \subset X$ be a rational tail (Example 0E3H). For any field extension K/k the base change $C_K \subset X_K$ is a finite disjoint union of rational tails.*

Date: stacks-project:curves.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Let $x \in C$ and $k' = \kappa(x)$ be as in the example. Observe that $C \cong \mathbf{P}^1_{k'}$ by Proposition 0C6U. Since k'/k is finite separable, we see that $k' \otimes_k K = K'_1 \times \ldots \times K'_n$ is a finite product of finite separable extensions K'_i/K. Set $C_i = \mathbf{P}^1_{K'_i}$ and denote $x_i \in C_i$ the inverse image of x. Then $C_K = \coprod C_i$ and $X'_K \cap C_i = x_i$ as desired. $\square$